

Project P10 - Toll Flow Anomaly Detection

ID: P10 | version: v1 | asset_type: project | language: en
sector: Transport / Infrastructure | duration: 10 months
stack: Apache Kafka, Databricks, Python, OpenCV

Client

Motorway operator J

Description

Anomaly detection on transaction streams and lane image classification.

Measurable result

Anomalies detected 3x faster

Context

Motorway operator collecting high-volume transaction and lane image data with delayed incident detection.

Challenges

Real-time volumes, noisy images and false-alert fatigue.

Solution

Kafka to Databricks streaming ingestion with statistical anomaly detection and a PyTorch lane image classifier.

Deliverables

- Streaming ingestion pipeline
- Anomaly detection service
- Lane image classifier
- Alert tuning guide

Team

- C09 - Nour El Houda Ben Salah - Streaming data engineer
- C13 - Thomas Girard - Computer vision engineer